

FORM 2 THE PATENTS ACT, 1970 (39 of 1970) & The Patent Rules, 2003 COMPLETE SPECIFICATION (See sections 10 & rule 13)											
1. TITLE OF THE INVENTION SWITCHED RELUCTANCE MOTOR (SRM) ARRANGEMENT WITH SINUSOIDAL INDUCTANCE PROFILE											
2. APPLICANT (S) <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 33%;">NAME</th> <th style="width: 33%;">NATIONALITY</th> <th style="width: 33%;">ADDRESS</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">Indian Institute of Technology Palakkad</td> <td style="text-align: center;">IN</td> <td style="text-align: center;">Nila campus, Kanjikode West, Palakkad, Kerala - 678623, India.</td> </tr> <tr> <td style="text-align: center;">IIT Palakkad Technology IHub Foundation</td> <td style="text-align: center;">IN</td> <td style="text-align: center;">Nila campus, Kanjikode West, Palakkad, Kerala - 678623, India.</td> </tr> </tbody> </table>			NAME	NATIONALITY	ADDRESS	Indian Institute of Technology Palakkad	IN	Nila campus, Kanjikode West, Palakkad, Kerala - 678623, India.	IIT Palakkad Technology IHub Foundation	IN	Nila campus, Kanjikode West, Palakkad, Kerala - 678623, India.
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3. PREAMBLE TO THE DESCRIPTION COMPLETE SPECIFICATION The following specification particularly describes the invention and the manner in which it is to be performed.											

TECHNICAL FIELD

[0001] The present disclosure relates to the field of electric motors. More particularly, the present disclosure relates to a switched reluctance motor (SRM) arrangement that creates a sinusoidal inductance profile.

BACKGROUND

[0002] Background description includes information that may be useful in understanding the present disclosure. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed disclosure, or that any publication specifically or implicitly referenced is prior art.

[0003] A radial or axial flux switched reluctance motor (SRM) is a type of electric motor characterized by the direction of the magnetic flux in relation to the axis of rotation. In a radial flux SRM, the magnetic flux flows radially (when crossing the airgap), meaning it travels perpendicularly from the rotor to the stator or vice versa. In an axial flux SRM, the magnetic flux flows parallel to the axis of rotation (when crossing the airgap). SRMs are provided with a laminated rotor to reduce eddy current losses. Conventional radial flux SRMs have the same laminate shape throughout the length of the axis. Moreover, the pole width and shape of conventional radial flux SRMs are fixed along the length of the axis typically resulting in trapezoidal inductance profiles. For conventional SRMs, the inductance plotted as a function of rotor angle look trapezoidal. Denoting θ as the electrical angle, there is a range of angle ($\Delta\theta$) in which the inductance is observed to increase thereby contributing to a positive torque (since $T \propto 0.5 \cdot i^2 \cdot dL/d\theta$, where $L(\theta)$ is the inductance of the phase, and 'i' is the current of the phase). Assuming a symmetric inductance profile, the maximum value of $\Delta\theta$ is 180 degrees. And for most of the trapezoidal profiles, $\Delta\theta$ will be much less than 180 degrees. Depending on the number of rotor and stator poles, the overlap of angle ($\delta\theta$) in which neighbour phases can contribute to positive torque is also a very small number. Further, a trapezoidal inductance profile leads to higher torque ripple because the overlap angle for positive $dL/d\theta$ of neighbouring phases is lesser, and necessitates faster

current injection/drop in them. Higher torque ripple results in more mechanical vibrations and noise, potentially reducing the smoothness of motor operation and affecting performance in several applications.

- 5 **[0004]** The torque ripple can generate higher acoustic noise levels. This can be a disadvantage in applications where quiet operation is critical, such as household appliances or electric vehicles. The sharper changes in current and voltage associated with a trapezoidal inductance profile can generate higher levels of EMI. Increased EMI can interfere with nearby electronic equipment and may require additional filtering and shielding, adding to the system's complexity and cost. The
- 10 trapezoidal inductance profile can lead to higher losses, particularly due to the increased harmonic content in the current waveform required to drive the motor. These harmonics cause additional core and copper losses. Reduced efficiency translates to higher energy consumption and more heat generation, necessitating more robust cooling mechanisms and potentially reducing the motor's lifespan.
- 15 **[0005]** To address these limitations, the present invention provides a novel switched reluctance motor arrangement that overcomes the shortcomings of the prior art.

OBJECTS OF THE PRESENT DISCLOSURE

- 20 **[0006]** Some of the objects of the present disclosure, which at least one embodiment herein satisfies are as listed herein below.

- [0007]** It is an object of the present disclosure to provide a switched reluctance motor (SRM) arrangement with a sinusoidal inductance profile, or a continuously varying inductance profile, that reduces the torque ripple which would minimize vibrations and noise, improving the overall smoothness and quietness of motor operation.
- 25

- [0008]** It is another object of the present disclosure to provide an SRM arrangement that creates a sinusoidal inductance profile, or a continuously varying inductance profile, which facilitates energy conversion by reducing harmonic content in the current waveform.
- 30

SUMMARY

[0009] This section is provided to introduce certain objects and aspects of the present disclosure in a simplified form that are further described below in the detailed description. This summary is not intended to identify the key features or the scope of the claimed subject matter.

[0010] The present disclosure relates to the field of electric motors. More particularly, the present disclosure relates to a switched reluctance motor (SRM) arrangement that creates a sinusoidal inductance profile, or a continuously varying inductance profile.

[0011] In an aspect of the present disclosure, a switched reluctance motor arrangement is disclosed. The switched reluctance motor (SRM) arrangement includes a plurality of stator poles and a plurality of rotor poles. A number of the plurality of rotor poles is obtained by subtracting two from N times a number of the plurality of stator poles. The SRM arrangement is provided with a rotor geometry, and a geometry of cross section of a rotor varies along an axis of the rotor for a radial flux motor, and along a radius of the rotor for an axial flux motor. Further, SRM arrangement is configured to create a sinusoidal inductance profile, or a continuously varying inductance profile.

BRIEF DESCRIPTION OF DRAWINGS

[0012] The accompanying drawings are included to provide a further understanding of the present disclosure and are incorporated in, and constitute a part of this specification. The drawings illustrate exemplary embodiments of the present disclosure, and together with the description, serve to explain the principles of the present disclosure.

[0013] In the figures, similar components, and/or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label with a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the

description applies to any one of the similar components having the same first reference label irrespective of the second reference label.

5 [0014] FIG. 1 illustrates an exemplary view of the proposed switched reluctance motor (SRM) arrangement for radial flux motor, in accordance with an embodiment of the present disclosure.

[0015] FIG. 2 illustrates an exemplary graphical representation of the sinusoidal inductance profile of the proposed SRM arrangement, in accordance with an embodiment of the present disclosure.

10 [0016] FIG. 3 illustrates exemplary representations depicting changes in rotor cross-section or rotor lamination shape (at different rotor depths) of the proposed SRM arrangement for radial flux motor, in accordance with an embodiment of the present disclosure.

15 [0017] FIG. 4 illustrates exemplary representations of the proposed SRM arrangement for a sinusoidal profile axial flux motor, in accordance with an embodiment of the present disclosure.

[0018] FIG. 5 illustrates an exemplary representation of a rotor with rotor laminates of the proposed radial flux SRM arrangement, in accordance with an embodiment of the present disclosure.

20 [0019] FIG. 6 illustrates an exemplary representation of rotor geometry of the proposed radial flux SRM arrangement, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

25 [0020] The following is a detailed description of embodiments of the disclosure depicted in the accompanying drawings. The embodiments are in such detail as to clearly communicate the disclosure. However, the amount of detail offered is not intended to limit the anticipated variations of embodiments. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit, and scope of the present disclosure as defined by the appended claims.

30 In the following description, numerous specific details are set forth in order to provide a thorough understanding of embodiments of the present invention. It will

be apparent to one skilled in the art that embodiments of the present invention may be practiced without some of these specific details.

[0021] Specific details are given in the following description to provide a thorough understanding of the embodiments. However, it will be understood by one of ordinary skill in the art that the embodiments may be practiced without these specific details. For example, circuits, systems, networks, processes, and other components may be shown as components in block diagram form in order not to obscure the embodiments in unnecessary detail. In other instances, well-known circuits, processes, algorithms, structures, and techniques may be shown without unnecessary detail to avoid obscuring the embodiments.

[0022] Also, it is noted that individual embodiments may be described as a process that is depicted as a flowchart, a flow diagram, a data flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed but could have additional steps not included in a figure. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc. When a process corresponds to a function, its termination can correspond to a return of the function to the calling function or the main function.

[0023] The word “exemplary” and/or “demonstrative” is used herein to mean serving as an example, instance, or illustration. For the avoidance of doubt, the subject matter disclosed herein is not limited by such examples. In addition, any aspect or design described herein as “exemplary” and/or “demonstrative” is not necessarily to be construed as preferred or advantageous over other aspects or designs, nor is it meant to preclude equivalent exemplary structures and techniques known to those of ordinary skill in the art. Furthermore, to the extent that the terms “includes,” “has,” “contains,” and other similar words are used in either the detailed description or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising” as an open transition word without precluding any additional or other elements.

[0024] Reference throughout this specification to “one embodiment” or “an embodiment” or “an instance” or “one instance” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present disclosure. Thus, the appearances of the
5 phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

[0025] The terminology used herein is for the purpose of describing particular
10 embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or
15 components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0026] The present disclosure relates to the field of electric motors. More
20 particularly, the present disclosure relates to a switched reluctance motor (SRM) arrangement that creates a sinusoidal inductance profile, or a continuously varying inductance profile.

[0027] In an embodiment of the present disclosure, a switched reluctance motor arrangement is disclosed. The SRM arrangement includes a plurality of stator poles
25 and a plurality of rotor poles. A number of the plurality of rotor poles equals two subtracted from N times a number of the plurality of stator poles, where N is a positive integer. The SRM arrangement is provided with rotor geometry, which is varying along an axis of a motor for radial flux and along a radius of the motor for axial flux. The rotor geometry or structure is configured to create a continuously
30 varying inductance profile (including sinusoidal). The geometry is such that the stator pole width equals the lowest rotor pole width among all depths of the rotor.

Also, the stator pole width equals the smallest distance between adjacent rotor pole edges among all depths of the rotor.

[0028] In an embodiment, the rotor geometry or structure is designed in a way that pole width and shape of the rotor varies along the axis of the rotor for radial flux motor and along the radius of the rotor for axial flux motor.

[0029] In an embodiment, the rotor geometry or structure is configured to maximize the overlap angle for torque sharing in neighbour phases.

[0030] In an embodiment, the rotor geometry or structure is configured to minimize torque ripple, acoustic noise, and vibrations.

[0031] In an embodiment, the stator pole width equals the lowest distance between edges of adjacent rotor poles among all the depths of the rotor (axial depth for radial flux motor and radial depth for axial flux motor).

[0032] In an embodiment, the rotor geometry or structure creates a sinusoidal inductance profile, or a continuously varying inductance profile for each electrical phase.

[0033] In an embodiment, the rotor geometry is realised by laminates of different shapes stacked together, or by machining of a single cylinder, or by 3D printing, or by moulding.

[0034] In an embodiment, the rotor geometry or structure creates a 180-degree electrical angle of rise of inductance and 180-degree electrical angle of fall of inductance.

[0035] In an embodiment, the rotor geometry creates an electrical angle of overlap, between neighbour phases having increasing inductance, given by “ $180 - (360/P)$ ” degrees, P being a number of electrical phases.

[0036] FIG. 1 illustrates an exemplary view of the proposed radial flux switched reluctance motor (SRM) arrangement, in accordance with an embodiment of the present disclosure.

[0037] As illustrated in FIG. 1, the radial flux switched reluctance motor (SRM) arrangement 100 operates based on the reluctance principle, which refers to a tendency of magnetic flux to follow the path of least magnetic resistance. The SRM arrangement 100 typically consists of a stator 102 with salient (protruding) poles,

each with concentrated windings. The SRM arrangement 100 further includes a rotor 104 that also has salient poles but does not contain any windings or permanent magnets. A stator core forms a stationary part of the motor and houses stator poles. The stator poles are the protruding parts of the stator core that hold the windings.

5 The number of stator poles is usually higher than the number of rotor poles. Copper coils are wound around the stator poles. When energized, the windings create a magnetic field that interacts with the rotor 104. The rotor core is designed to rotate within the stator 102. Alternatively, an outrunner option is also possible. The rotor core has salient poles (protruding parts) but no windings or permanent magnets.

10 **[0038]** In an embodiment of the present disclosure, the salient poles of the rotor 104 align with the stator poles to minimize reluctance and produce torque. There is a small air gap between the stator 102 and the rotor 104. The size and uniformity of the air gap are crucial for the performance of the motor, affecting the magnetic flux and inductance. The operation of the SRM arrangement 100 is based on the
15 principle of reluctance torque, which is generated by a tendency of the rotor 104 to move to a position where the reluctance (magnetic resistance) is minimized. This is achieved by sequentially energizing the stator windings in a specific order, creating a rotating magnetic field that pulls the rotor 104 into alignment with the energized stator poles.

20 **[0039]** In an embodiment of the present disclosure, a number of the plurality of rotor poles equals two subtracted from N times a number of the plurality of stator poles, where N is a positive integer. The rotor geometry or structure creates a sinusoidal inductance profile, or a continuously varying inductance profile for each electrical phase. For a radial flux SRM (100), the rotor geometry is such that the
25 stator pole width is made equal to a lowest rotor pole width among all depths of the rotor 104 along a rotational axis 106 of the rotor 104 in order to create a sinusoidal inductance profile.

[0040] In an embodiment of the present disclosure, the sinusoidal inductance profile in the SRM arrangement 100 may be achieved by designing a rotor geometry
30 in such a way that the inductance of the stator windings changes sinusoidally as the rotor rotates. This is desirable for smoother torque production and lower acoustic

noise. The rotor geometry is designed to vary along the rotor axis of SRM arrangement 100 for radial flux motor. The rotor geometry is configured to minimize torque ripple, acoustic noise, and vibrations.

[0041] In an embodiment of the present disclosure, the rotor laminations are designed with specific varying geometric shapes that modulate the effective overlap area between the rotor and the stator poles in a sinusoidal manner, or a continuous manner. The laminations are stacked to form the complete rotor 104. Further, the rotor geometry is designed in a way that pole width and shape of the rotor 104 varies along a length of the axis, for radial flux. This modulation of effective overlap area results in the continuously varying inductance profile. As described herein “continuously varying inductance profile” refers to a profile which neither has a flat bottom nor a flat top. Further, the inductance value will rise monotonically from lowest to highest value, and decrease monotonically from the highest to lowest value. The derivative $dL/d\theta$ will be zero only at two values of θ (at maxima and minima).

[0042] FIG. 2 illustrates an exemplary graphical representation of the sinusoidal inductance profile of the proposed SRM arrangement, in accordance with an embodiment of the present disclosure.

[0043] As illustrated in FIG. 2, a sinusoidal inductance profile 200 in a switched reluctance motor (SRM) refers to a design where the inductance of the stator windings of the SRM arrangement 100 varies in a sinusoidal manner as the rotor rotates. This means that the inductance changes smoothly and periodically, resembling a sine wave, as a function of the rotor position. Achieving the sinusoidal inductance profile 200 is desirable for smoother torque production and reduction of torque ripple and acoustic noise.

[0044] In an embodiment of the present disclosure, the rotor geometry of the SRM arrangement 100 makes the inductance profile resemble a sinusoidal waveform, with $\Delta\theta$ as 180 degrees, and $\delta\theta$ as “ $180-(360/P)$ ” degree, where $P \geq 3$ is the no. of electrical phases. Since two neighbour phases have positive $dL/d\theta$ for a large range of angle $\delta\theta$, it is easier to build-up current in the incoming phase and bring down the current in outgoing phase in a smoother manner (for a given same

revolutions per minute (RPM) speed). It is much easier to achieve elimination of torque ripple due to smoother phase commutation. To achieve the sinusoidal inductance profile 200, the rotor laminations of the SRM arrangement 100 are designed in a way that the effective overlap area between the rotor 104 and the stator poles vary in a sinusoidal manner, or a continuous manner. The excitation of the stator windings must be synchronized with the rotor position to take advantage of the sinusoidal inductance variation. This means controlling the current in the windings to align with the varying inductance.

[0045] In an embodiment of the present disclosure, in a sinusoidal inductance profile, the inductance L of a stator winding varies according to the equation: $L(\theta)=L_0+L_1 \cos(\theta)$, where $L(\theta)$ is the inductance as a function of electrical rotor angle θ , L_0 is the average inductance, L_1 is the amplitude of inductance variation. The sinusoidal inductance profile 200 in the SRM arrangement 100 enhances the performance of the motor by providing smoother torque production and reducing undesirable effects like torque ripple and noise.

[0046] FIG. 3 illustrates exemplary representations depicting changes in rotor cross-section or rotor lamination shape (along the depth of the axis) of the proposed radial flux SRM arrangement (100), in accordance with an embodiment of the present disclosure.

[0047] As illustrated in the representations 300 of FIG. 3, the shape of the cross-section or laminations of the rotor changes along the length of the axis of the rotor to create the sinusoidal inductance profile 200. The figure depicts depths 0-5 from the same rotor position as can be understood from the cross-section views of the SRM arrangement 100 taken at different depths along the rotational axis (106). Phase C has the highest inductance, since there is a fully aligned rotor pole at all depths. Phase A has the lowest inductance, since there is no rotor pair aligned in all depths. However, if the rotor 104 rotates even by a small angle in a clockwise or anticlockwise direction, then phase A inductance will increase because at Depth 0 there are rotor pole areas or rotor laminations coming into overlap with stator poles. Hence, the continuously varying sinusoidal inductance profile 200 is created by the SRM arrangement 100, which makes $\Delta\theta$ as 180 degrees. Thus, for continuous

inductance variation (with neither flat bottom nor flat top), at least two conditions need to be met: the stator pole width should be equal to the lowest rotor pole width for all depths. Further, the stator pole width should be equal to 'd', where 'd' is the lowest distance between edges of the adjacent rotor poles among all depths.

5 Depending on the shape and width of rotor pole at different depths, the SRM arrangement 100 may achieve different profiles for $L(\theta)$ other than sinusoidal inductance profile.

[0048] FIG. 4 illustrates exemplary representations of the proposed SRM arrangement for a sinusoidal profile axial flux motor, in accordance with an
10 embodiment of the present disclosure. It has the continuously varying sinusoidal inductance profile 200 by variation of rotor pole width in the radial direction.

[0049] As illustrated in FIG. 4, both radial flux 100 and axial flux 400 SRM arrangements include the salient stator and the rotor poles. The stator poles and the rotor poles are related in a way that a number of Rotor Poles (R) equals (N times the number of stator poles (S)) minus 2, or $R=N*S-2$, where $S \geq 6$ (assume only
15 even S), and the integer $N \geq 1$. With multiplicity of poles per phase, $M*S$ is the no. of stator poles, $M*R$ is the no. of rotor poles, where $M \geq 1$ is the multiplicity integer. The number of electrical phases (P) is $S/2$ (assume only even S). For $N=1$, the commutation frequency is lower as compared to higher values of N. At the same
20 time, the continuously varying sinusoidal inductance profile 200 is also achieved through changes in rotor geometry even for $N=1$.

[0050] In an embodiment of the present disclosure, the rotor geometry may have different shapes at different depths of the rotor 104, including axial depth for radial flux motor and radial depth for axial flux motor. The rotor teeth size, teeth
25 depth and profile angles may vary. The inductance profile of any motor phase of the SRM arrangements 100 and 400 will have continuously varying inductance with neither a flat bottom nor a flat top. The electrical angle of the rise of inductance will be 180 degrees while the electrical angle of the fall of inductance would be 180 degrees. The sinusoidal inductance profile 200 is obtained for each electrical phase
30 of the SRM arrangements 100 and 400. The electrical angle of overlap, for the continuously varying inductance profile, between neighbour phases of the SRM

arrangements 100 and 400 having increasing inductance is given by “ $180-(360/P)$ ” degrees. In order to achieve this the stator pole width has to be equal to the lowest rotor pole width among all depths and the stator pole width may be equal to ‘d’, where ‘d’ is the lowest distance between edges of the adjacent rotor poles among all depths (axial depth for a radial flux motor and radial depth for an axial flux motor). The inductance profile of the SRM arrangements 100 and 400 may have a flat top or bottom, if desired, by varying the rotor geometry between a standard inductance profile and the proposed sinusoidal inductance profile.

[0051] In an example embodiment, the SRM arrangements (100 and 400) that creates the sinusoidal or continuously varying inductance profile 200 is used in electric vehicles (EVs). Traditional SRMs are known for high torque ripple, which can cause vibrations and noise. By designing the motor to have the sinusoidal or continuously varying inductance profile 200, the torque ripple may be significantly reduced, leading to smoother and quieter operation. The reduced torque ripple and smoother operation enhances the overall comfort of the vehicle, making the driving experience more pleasant for passengers. Lower torque ripple and smoother operation reduce the noise generated by the motor, contributing to a quieter cabin environment. The improved torque characteristics and smoother operation reduce mechanical stress on the motor components, potentially increasing the durability and reliability of the motor.

[0052] FIG. 5 illustrates an exemplary representation of a rotor with rotor laminates of the proposed radial flux SRM arrangement, in accordance with an embodiment of the present disclosure.

[0053] As illustrated in FIG. 5 the rotor 104 of the SRM arrangement 100 is typically made up of a stack of thin steel laminations, hereafter referred to as rotor laminations 500. The rotor laminations 500 are insulated from each other to minimize eddy current losses. The rotor 104 does not have any windings or permanent magnets but has a salient pole structure that relies purely on the variation in reluctance. The rotor laminations 500 are punched or laser-cut into specific shapes that define the rotor poles. The sinusoidal inductance profile 200 means that the inductance varies smoothly in a sinusoidal manner as the rotor 104 moves, rather

than abruptly changing between high and low inductance states. This smooth variation is preferred because it reduces torque ripple and noise, leading to quieter motor operation, and reduced electromagnetic interference.

5 **[0054]** FIG. 6 illustrates an exemplary representation of rotor geometry of the proposed SRM arrangement, in accordance with an embodiment of the present disclosure.

10 **[0055]** Illustrated in FIG. 6 is a representation 600 of the rotor geometry of the radial flux SRM arrangement 100. The rotor geometry features an increasing and decreasing pole width. This is an alternative to monotonic increase of pole width from one end of the axis to another, and results in a continuously varying inductance profile (200).

15 **[0056]** While the foregoing describes various embodiments of the invention, other and further embodiments of the invention may be devised without departing from the basic scope thereof. The scope of the invention is determined by the claims that follow. The invention is not limited to the described embodiments, versions or examples, which are comprised to enable a person having ordinary skill in the art to make and use the invention when combined with information and knowledge available to the person having ordinary skill in the art.

20 **ADVANTAGES OF THE INVENTION**

[0057] The present disclosure provides a switched reluctance motor (SRM) arrangement with a sinusoidal inductance profile that reduces the torque ripple which minimizes vibrations and noise, improving the overall smoothness and quietness of motor operation.

25 **[0058]** The present disclosure provides an SRM arrangement that creates a sinusoidal inductance profile which facilitates energy conversion by reducing harmonic content in the current waveform.

We Claim:

1. A switched reluctance motor (SRM) arrangements (100, 400) comprising a plurality of stator poles and a plurality of rotor poles, a number of the plurality of rotor poles being two subtracted from N times a number of the plurality of stator poles, wherein the SRM arrangement (100, 400) are provided with a rotor geometry, and a geometry of cross section of a rotor (104) varies along an axis of the rotor (104) for a radial flux motor (100), and along a radius of the rotor (104) for an axial flux motor (400), and is configured to create a sinusoidal inductance profile (200), or a continuously varying inductance profile.
2. The SRM arrangement (100) as claimed in claim 1, wherein the rotor geometry is realised by rotor laminations (500) of different shapes by stacking the rotor laminations (500) together, by machining of a single cylinder, by 3D printing techniques, or by moulding.
3. The SRM arrangement (100) as claimed in claim 2, wherein the rotor laminations (500) are designed in a way that pole width and shape of the rotor (104) varies along a length of the rotor axis for the radial flux motor (100).
4. The SRM arrangement (100) as claimed in claim 2, wherein the rotor geometry is configured to maximize an overlap angle for torque sharing in neighbour phases.
5. The SRM arrangement (100) as claimed in claim 2, wherein the rotor geometry is configured to minimize torque ripple, acoustic noise, and vibrations.
6. The SRM arrangement (100) as claimed in claim 1, wherein the stator pole width equals a lowest distance between edges of adjacent rotor poles among all the depths along a rotational axis (106) of the rotor (104) and also, equals a lowest rotor pole width among all depths along a rotational axis (106) of the rotor (104).
7. The SRM arrangement (100) as claimed in claim 1, wherein the rotor geometry creates the sinusoidal inductance profile (200), or a continuously varying inductance profile for each electrical phase.

8. The SRM arrangement (100) as claimed in claim 1, wherein the rotor geometry creates a 180-degree electrical angle of rise of inductance and 180-degree electrical angle of fall of inductance.

9. The SRM arrangement (100) as claimed in claim 1, wherein the rotor geometry creates an electrical angle of overlap, between neighbour phases having increasing inductance, given by “ $180-(360/P)$ ” degrees, P being a number of electrical phases.

**For Indian Institute of Technology Palakkad and
IIT Palakkad Technology IHub Foundation**



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ABSTRACT

SWITCHED RELUCTANCE MOTOR (SRM) ARRANGEMENT WITH SINUSOIDAL INDUCTANCE PROFILE

5 Embodiments of the present disclosure relate to a switched reluctance motor (SRM)
arrangement (100). The SRM arrangement (100) includes a plurality of stator poles
and a plurality of rotor poles. A number of the plurality of rotor poles equals two
subtracted from N times the number of the plurality of stator poles (N is a positive
integer). A stator pole width equals to the lowest rotor pole width among all depths
10 of a rotor, also, the stator pole width equals to the lowest distance between adjacent
rotor pole edges among all depths. The SRM arrangements (100 and 400) are
provided with a rotor geometry, where the cross section of the rotor (104) is varying
along the axis of the rotor for radial flux motor (100) and along the radius of the
rotor for axial flux motor (400). The rotor geometry is configured to create a
15 sinusoidal inductance profile (200).

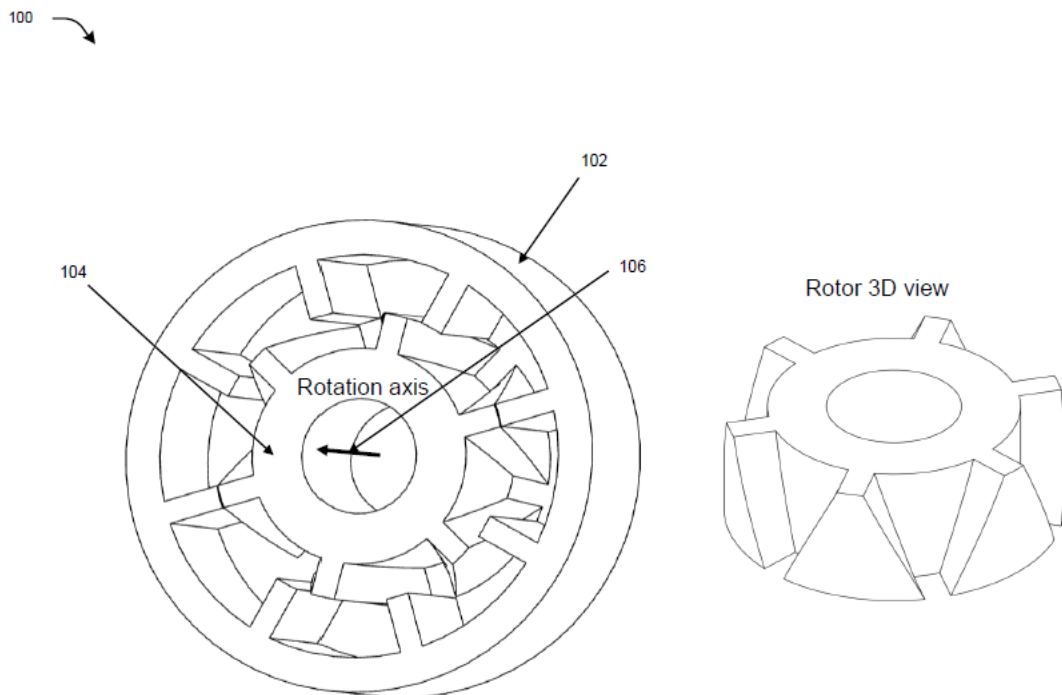


FIG. 1